

University of Central Punjab
Faculty of Information Technology and Computer Science
INFORMATION SECURITY (LAB)
Week 1 - Introductory Lab Manual

1. Course Information

Course Title	Information Security (Lab)
Credit Hours	1 CH
Course Coordinator	Dr Ghulam Mustafa
Lab Instructor	Syeda Mehak Zehra
Course Tools	Nmap, Wireshark, UFW etc. in Kali Linux

2. Assessment Weightages

Final Lab Exam	40%
Lab Midterm	25%
Graded Labs	25%
Class Participation	10%
Lab Total	100%

3. Course Introduction

Information Security introduces the fundamental concepts used to protect information and computing systems. The lab emphasizes practical application of security techniques, including cryptography, traffic analysis, system security, scanning, and common attack/defense concepts.

At the beginning of the course, students will first become familiar with classical encryption ideas before moving toward modern security tools and techniques.

4. Week 1 Lab: Introduction to Classical Ciphers

The purpose of Week 1 is primarily introductory. Students should understand what encryption is, why a key is used, and how classical substitution ciphers transform plaintext into ciphertext.

5. Basic Terminology

Plaintext	Original readable message.
Ciphertext	Encrypted or transformed message.
Encryption	Process of converting plaintext into ciphertext.
Decryption	Process of recovering plaintext from ciphertext.
Key	A value used to control the encryption/decryption process.

6. Classical Ciphers - Brief Overview

Caesar Cipher: A substitution cipher in which each alphabetic character is shifted by a fixed number of positions. It is the main cipher demonstrated in this introductory lab.

Vigenère Cipher: A polyalphabetic cipher that uses a keyword so that the shift can vary from one character to another.

Auto-Key Cipher: A variation of the Vigenère idea in which the plaintext is used to extend the initial key.

Playfair Cipher: A digraph substitution cipher that encrypts pairs of letters using a 5 x 5 key matrix.

1. Caesar Cipher

In Caesar Cipher, every letter is moved by the same fixed shift. For example, with a key of 3:

A -> D B -> E C -> F ... X -> A Y -> B Z -> C

Encryption formula: $C = (P + K) \bmod 26$

Decryption formula: $P = (C - K) \bmod 26$

```

2  #include <iostream>
3  using namespace std;
4
5  int main()
6  {
7      string text;
8      int key;
9      cout << "Enter text: ";
10     cin >> text;
11
12     cout << "Enter key: ";
13     cin >> key;
14
15     for (int i = 0; i < text.length(); i++)
16     {
17         text[i] = (text[i] - 'A' + key) % 26 + 'A';
18     }
19
20     cout << "Encrypted text: " << text;
21     return 0;
22 }

```

Practice Task:

Implement encryption and decryption functions.
 Test with key=3 on plaintext student's full name.
 Values must not be hardcoded.

2. Vigenere Cipher

A polyalphabetic substitution cipher that uses a keyword to vary the shift amount for each alphabetic character.

Encryption: $C_i = (P_i + K_i) \bmod 26$

Decryption: $P_i = (C_i - K_i) \bmod 26$

plaintext HELLOWORLD

key KEYKEYKEYKE

```

3
4  int main() {
5
6      string text, key;
7      cout << "Enter plaintext: ";
8      cin >> text;
9      cout << "Enter keyword: ";
10     cin >> key;
11
12     for (int i = 0; i < text.length(); i++)
13     {
14         int p = text[i] - 'A';
15         int k = key[i % key.length()] - 'A';
16
17         text[i] = (p + k) % 26 + 'A';
18     }
19
20     cout << "Encrypted text: " << text << endl;
21
22     return 0;
23 }

```

Practice Task:

- * Implement encryption and decryption functions.
- * Test with keyword “KEY” on plaintext “CRYPTOGRAPHY”.
- * Values must not be hardcoded.

3. Auto-Key Cipher

A modification of Vigenère Cipher where the plaintext itself becomes part of the key after the initial keyword.

```

plaintext HELLOWORLD
key          KEYHELLOWO

```

Pseudo Code:

INPUT: Plaintext, Initial_Key

GENERATE extended_key = Initial_Key + first (len(Plaintext) - len(Initial_Key)) letters of Plaintext

FOR each character:

Cipher = (Plain + Key) mod 26

OUTPUT: Ciphertext

Practice Task:

- * Implement encryption and decryption functions.
- * Test with keyword “KEY” on plaintext “CRYPTOGRAPHY”.
- * Values must not be hardcoded.